Lab 1 | Selected Report Excerpts

A compilation of original excerpts from different students' reports.

This document combines selected passages from four student reports.
These excerpts are not complete model answers.

Source guide
Steps 1 and 4: Nathan Gabrielle. Steps 2, 3, 5 and 6: Jiang Pinheng.
Step 7: Nathan Gabrielle (selected comparison and results).
Step 8, Q1: Jiang Pinheng. Q2: Zhu Zhexian and Ni Zhangzhe.
Step 8, Q3: Zhu Zhexian (definitions) and Jiang Pinheng (observed cases).

Step 1 | Dataset 1

Nathan Gabrielle | Original page 1

II. Step 1: Dataset 1 / Model 1

Class	Number of Uploaded Samples
Mandatory	100
Prohibitory	100
Warning	100

Table 1: Dataset 1 class and uploaded samples.

Description:

Dataset 1 is collected through Google. We collected 100 samples for each class, for a total of 300 samples. All these images are flat 2D stock vector illustrations of traffic signs, centered in the middle of the image, with a plain one-color background.

Step 2: Collect a Fixed External Test Dataset

A fixed external test dataset was collected independently of the uploaded samples: 20 additional images per class (60 images in total). These images were never uploaded to any class in Teachable Machine and were not used for training in any way. The external test images deliberately introduce conditions that differ from Dataset 1: a different indoor location (a white-walled room and a corridor with wooden doors and artwork), a wider range of distances (long shots to close-ups), different camera angles (frontal, three-quarter, side-profile and rear views) and additional objects (a tablet held by the participant or placed on the floor). Table 2 records the number of external test samples per class, and Table 3 describes how many samples represent each collection condition.

Table 2: External test dataset – number of samples per class (identical class sizes).

Class	External test samples
Standing	20
Walking	20
Fallen	20
Total	60

Table 3: External test dataset – samples per collection condition (samples numbered in file order).

Class	Condition group	Samples	Representative features
Standing	Samples 1–5	5	White-walled room, frontal view (samples 1–3 distant; 4–5 close-up)
Standing	Sample 6	1	Rear view (participant facing away)
Standing	Samples 7–9	3	Corridor, three-quarter / side pose, no prop
Standing	Samples 10–19	10	Corridor, front-facing pose holding a tablet
Standing	Sample 20	1	Corridor, three-quarter pose holding a tablet
Walking	Samples 1–10	10	Corridor with artwork wall, mid-stride, side profile
Walking	Samples 11–15	5	Corridor near service door / signage, mid-stride
Walking	Samples 16–20	5	Glass double-door area, mid-stride
Fallen	Samples 1–3	3	Crawling away from the camera, no props
Fallen	Samples 4–14	11	Kneeling / crawling near the planting and recycling corner
Fallen	Samples 15–20	6	Kneeling on the floor, tablet placed on the floor nearby

Step 3: Train and Evaluate Model 1

Model 1 was trained in Teachable Machine using Dataset 1 with the following Advanced settings: 50 epochs, batch size 16 and learning rate 0.001. Teachable Machine reported an internal test accuracy of 100% on its own 15% split. The trained project was saved as Model1.tm. The model was then evaluated with every sample of the fixed external test dataset using the Preview panel: each of the 60 images was shown once and its predicted class was compared with the true class. Table 4 records the configuration and the results, Table 5 gives the per-class results, and equation (1) shows the external test accuracy calculation as defined in the lab handout.

Table 4: Model 1 – training configuration and evaluation results.

Item	Value
Dataset size (uploaded samples)	300 images (100 per class)
Fixed external test dataset size	60 images (20 per class)
Epochs	50
Batch size	16
Learning rate	0.001
Correct predictions (external test)	23 of 60
Internal test accuracy (Teachable Machine)	100% (45/45 internal test samples; 15/15 per class)
External test accuracy	38.33% (see equation (1))

External test accuracy = (Number of correct predictions / Total number of external test samples) × 100% (1)

Model 1: 23 / 60 × 100% = 38.33%.

Table 5: Model 1 – external test results per class (20 test samples per class).

Class	Correct / total	Accuracy	Misclassified samples
Standing	0 / 20	0%	All samples 1–20
Walking	20 / 20	100%	None
Fallen	3 / 20	15%	Samples 4–20 (only samples 1–3 correct)
Overall	23 / 60	38.33%	—

V. Step 4: Dataset 2 / Model 2

Class	Number of Uploaded Samples
Mandatory	100
Prohibitory	100
Warning	100

Table 4: Dataset 2 class and uploaded samples.

Description:

Dataset 2 is collected through the “CCTSDB” dataset. We selected 100 samples for each class, for a total of 300 samples from the 1000-1999 range. All these images are cropped so that the sign is clearly visible and zoomed in. Approximately 90% of the images have the sign centered, while the rest have it positioned on the sides. These images were taken in a real environment (with mixed sunny and rainy conditions), so they have a real world background.

Step 5: Train and Evaluate Model 2

Model 2 was trained in Teachable Machine using Dataset 2 with exactly the same settings as Model 1 (50 epochs, batch size 16, learning rate 0.001). Teachable Machine reported an internal test accuracy of 95.6% (per-class accuracies: Standing 0.87 = 13/15, Walking 1.00, Fallen 1.00; 43 of 45 internal test samples correct). The trained project was saved as Model2.tm. The same fixed external test dataset of 60 images was then evaluated once with the Preview panel. Table 7 records the configuration and results and Table 8 gives the per-class results.

Table 7: Model 2 – training configuration and evaluation results.

Item	Value
Dataset size (uploaded samples)	300 images (100 per class)
Fixed external test dataset size	60 images (20 per class)
Epochs	50
Batch size	16
Learning rate	0.001
Correct predictions (external test)	49 of 60
Internal test accuracy (Teachable Machine)	95.6% (43/45 internal test samples; Standing 13/15, Walking 15/15, Fallen 15/15)
External test accuracy	81.67% (49 / 60 × 100%)

Table 8: Model 2 – external test results per class (20 test samples per class).

Class	Correct / total	Accuracy	Notes
Standing	10 / 20	50%	Correct for samples 4–6 and 13–19; incorrect for 1–3, 7–12 and 20
Walking	19 / 20	95%	Incorrect only for sample 19
Fallen	20 / 20	100%	All samples correct
Overall	49 / 60	81.67%	—

Step 6 | Additional experiment

Jiang Pinheng | Original page 8 | Data composition, settings and evaluation

Reporting example; the boundary of the 'Fallen' class requires clarification.

Step 6: Additional Experiment – Combined Dataset (Bonus)

An additional experiment was conducted to test how the composition of the uploaded data affects external performance. Dataset 3 combines the two collection strategies in equal parts: for every class, the first 50 photos of Dataset 1 were merged with the first 50 newly captured photos of Dataset 2 (100 uploaded samples per class; class definitions, training parameters and the fixed external test dataset unchanged). Table 10 records the composition. Model 3 was trained on Dataset 3 with the same settings as the other models (50 epochs, batch size 16, learning rate 0.001) and evaluated once on all 60 images of the same external test dataset; the trained project was saved as Model3.tm (included in the projects folder). Tables 11 and 12 record the results.

Table 10: Dataset 3 (bonus experiment) – composition of the combined dataset.

Class	Uploaded samples	Source of the samples
Standing	100	50 photos from Dataset 1 (same corridor, frontal) + 50 newly captured photos of Dataset 2 (second participant, rear and side views, tablet poses)
Walking	100	50 photos from Dataset 1 (frontal approach) + 50 newly captured photos of Dataset 2 (other directions, locations and distances)
Fallen	100	50 photos from Dataset 1 (prone, fixed location) + 50 newly captured photos of Dataset 2 (kneeling/crawling, several locations, with tablet)
Total	300	—

Table 11: Model 3 – training configuration and evaluation results.

Item	Value
Dataset size (uploaded samples)	300 images (100 per class)
Fixed external test dataset size	60 images (20 per class)
Epochs	50
Batch size	16
Learning rate	0.001
Correct predictions (external test)	46 of 60
Internal test accuracy (Teachable Machine)	100% (45/45 internal test samples; 15/15 per class)
External test accuracy	76.67% (46 / 60 × 100%)

Table 12: Model 3 – external test results per class (20 test samples per class).

Class	Correct / total	Accuracy	Notes
Standing	7 / 20	35%	Correct for samples 6 and 14–19; incorrect for 1–5, 7–13 and 20
Walking	19 / 20	95%	Incorrect only for sample 19
Fallen	20 / 20	100%	All samples correct
Overall	46 / 60	76.67%	—

Step 7 | Model comparison: selected results

Nathan Gabrielle | Original page 4 | Table, figure and accompanying description

The original report labels this section "Step 8"; that heading is retained below.

IX. Step 8: Compare and Analyze the Models

Model	Dataset Size	External Test Size	Epochs	Batch Size	Learning Rate	Correct Predictions	Internal Test Accuracy	External Test Accuracy
Model 1	300	60	50	16	0,001	40	100%	66,7%
Model 2	300	60	50	16	0,001	51	100%	85%
Model 3	300	60	50	16	0,001	44	100%	73,3%

Table 8: Comparison of all models

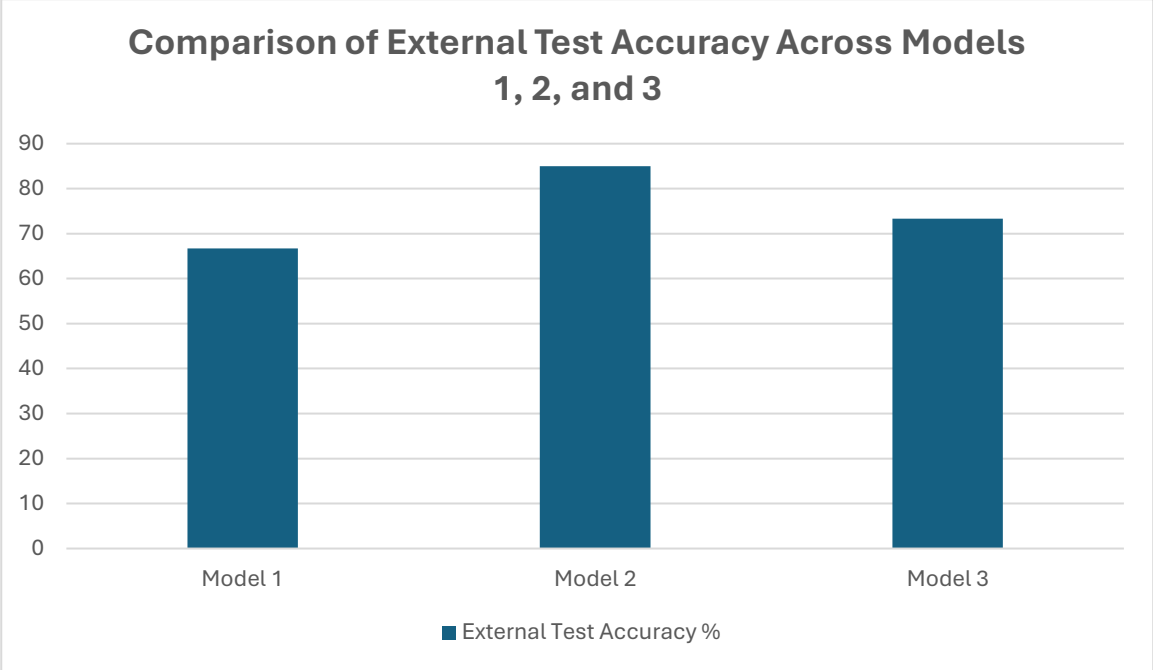


Figure 1: Comparison of external test accuracy (%) across Models 1, 2, and 3

Correctly Classified External Test Samples

Model 2 achieved the highest external test accuracy (85%, 51/60 correct), and its correctly classified samples were those captured under clear, and well lit conditions. The external test dataset (Table 2) was drawn from the CCTSDB dataset and cropped to a 1:1 ratio, with approximately 90% of signs centred and 10% positioned toward the sides. Model 2 correctly classified 19/20 mandatory, 19/20 prohibitory, and 13/20 warning samples.

Step 8 | Questions 1 and 2

Selected conceptual explanations from three reports

Q1 | Controlled comparison

Jiang Pinheng | Original page 13 | Opening explanation only

Question 1 — Why must the class definitions, number of uploaded samples, training parameters and fixed external test dataset remain unchanged when comparing Model 1 and Model 2?

Because the comparison should be a controlled experiment in which the only varying factor is the dataset (i.e. the data-collection strategy). If class definitions changed, the two models would solve different tasks; if the number of uploaded samples changed, the comparison would also reflect a change in data volume; if the epochs, batch size or learning rate changed, the comparison would reflect a change in optimisation rather than data; and if the external test dataset changed, the accuracy values would measure performance on test sets of different difficulty and would no longer be comparable.

Q2 | How a test set influences model development

Zhu Zhexian | Original page 9 | Opening explanation only

Answer: No, because human judgement and intervention are involved. The redesigned training datasets incorporate information gained from examining the test set, so the test set can no longer be considered completely unseen.

Q2 | Why an untouched test set helps

Ni Zhangzhe | Original page 2 | Explanation of independent evaluation

An additional untouched test dataset provides stronger evidence because neither the model nor the dataset-design decisions have been adjusted based on its results. Evaluating the final model on this new set gives a more independent estimate of how well it may classify genuinely new, real-world images.

Step 8 | Question 3

Confidence, accuracy and observed high-confidence errors

Conceptual distinction

Zhu Zhexian | Original page 9 | Definition excerpt

> 3. Does a higher confidence score always indicate better model performance? Explain the difference between confidence and accuracy, and support your answer with at least one result from Model 1 or Model 2.
>
> Answer: A higher confidence score does not always indicate better model performance. Confidence is an internal prediction score that reflects how certain a trained AI model is about its prediction for a particular test sample, but it does not guarantee that the prediction is correct. Accuracy, by contrast, measures the proportion of correct predictions across the test dataset rather than how confident those predictions are.

Concrete cases from a different experiment

Jiang Pinheng | Original page 14 | High-confidence errors only

> Two results from this lab show that high confidence does not imply good performance. First, Model 1 returned 'Walking' with 100% confidence for a standing test image — despite zero of its 20 standing predictions being correct — and Model 2 returned 'Standing' with 96% confidence for the single walking sample it misclassified (sample 19), and Model 3 gave the same high-confidence answer as Model 1 on the same standing sample (walking, 100%). In all of these cases the models were confidently wrong.